PROVEIT RESEARCH CONTINUATION

Relative Top-Handle Elimination

and the Limits of Forest Optimization

in Compressed Unknot Recognition

Universal source trees, homological constraints,
exceptional-row kernels, and independently replayed certificates

Prepared for Vladimir Reshetnikov and the ProveIt project

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Abstract. We remove a concrete unsupported case from ProveIt's compressed normal-cut pipeline. Every three-handle inherited from an interior source vertex can be eliminated by source-authenticated two/three-handle cancellations, relative to the recorded parallelity interfaces. One boundary-rooted tree of the source one-skeleton specializes to a valid schedule for every admissible normal surface, including the empty surface and surfaces with arbitrarily large binary coordinates. The scheduling work is linear in the bounded core size; the additional trace has $O(t \log(t+2))$ bits.

We then examine whether a better top-handle forest can make the remaining one/two-handle cancellations succeed. For abstract short occurrence rows this is a two-graphic-matroid partition problem with an exact rank-deficiency certificate. A further homological argument substantially narrows its geometric significance: the boundary-of-boundary identity makes the residual short-row nullity independent of the forest on the genuine geometric subclass. We also give an exact contraction kernel for a fixed forest with arbitrary exceptional rows, isolating the part that requires more than graphic matroid methods.

The native integration increases independently certified whole-cut censuses from 85 to 92 on the retained 106-case corpus, preserving all earlier successes. On a family of genuine subdivided solid tori, the largest measured case gives a 28.8-fold scheduling gain and a 1.94-fold gain when common source-incidence reconstruction is included. Small cases regress. These are local advances: a complete quasipolynomial unknot-recognition bound is not established. The article gives detailed proofs, evidence, integration contracts, and a research programme aimed at the remaining geometric and global complexity obligations.

Based on repository commit `8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`.

All performance claims refer to the supplied reproducible experiments. Mathematical generality, native implementation scope, and unproved proposals are stated separately.

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1 Position of the result

The immediate outcome is a certified extension of a supplied-normal-cut primitive. ProveIt's original compressed source construction already retained the exceptional handles, their attaching charts, and the parallelity interfaces. Its terminal handlebody consumer refused any core containing a three-handle. A diagnostic script demonstrated promising deletions, but did not provide independent source replay. We prove the relative geometric fact needed for those deletions, reconstruct the missing cap incidences from the source, and compose the resulting proof with the existing terminal certificates.

This is consequential because a source interior vertex is an artefact of the triangulation, not evidence of difficult knot topology. A recognizer should not lose its compressed terminal operation merely because such a vertex was introduced. The universal source-tree theorem shows that this obstruction can be removed uniformly, without searching normal-coordinate supports or expanding repeated sheets.

1.1 The quasipolynomial objective and its present status

For an input knot diagram with n crossings, the intended target is a deterministic bound of the form

$$\exp\big((\log(n+2))^{O(1)}\big),$$

with $n^{O(\log n)}$ being the particularly strong form announced in 2021. The announcement and a proved, implemented end-to-end complexity theorem must be distinguished. Lackenby's July 2026 preprint presents an algorithm using incompressible surfaces, boundary patterns, and hierarchies. Its Section 9 explicitly withholds a running-time estimate for insufficiently specified steps. Proposition 9.1 bounds the number of steps by $L(g+1)^L$, in the paper's notation; Proposition 10.2 bounds a refined partial hierarchy length by $4c_q(\mathcal{H})$. A linear hierarchy bound is not a logarithmic depth bound. Polynomial encoding and verification of an essential hierarchy in Theorem 14.1 do not give a polynomial procedure for discovering it [1, 2].

Our mathematical target is therefore a carefully delimited part of that programme. A bounded compressed core must support inexpensive geometric operations with enough source information to justify their use after a cut. The present top-handle operation meets that requirement for the native parallelity interfaces. It does not supply arbitrary inherited-pattern transport, a complete useful-surface search, or a global recurrence bounding all intermediate representations.

1.2 What is reused and what is contributed

Removing three-handles from a compact connected three-manifold with nonempty boundary is classical. Jackson gives the needed unmarked statement and a dual-path proof in Lemma 3.2 of [3]. The same paper explains why recovering parallelity-bundle topology is weaker than retaining its inclusion in the ambient manifold. Our contribution is the source-specific relative construction, its reusable specialization across all normal cuts, and its independently verified native implementation. We do not claim a new general existence theorem for three-handle elimination.

The graph-matroid partition theorem is also classical [4]. Its application here gives a precise abstract optimization problem and certificates of optimal residual deficiency. The chain-complex constraint in section 5, however, shows why abstract successes cannot automatically be interpreted

as geometric performance gains. This limitation is part of the research result, not an omitted qualification.

Within ProveIt, the source-parallelity construction, the initial uncut two/three-handle queue, the stopping-set characterization of positive raw cancellation, and the saturated-bundle projection are prior work. The ten incoming packages were audited for these overlaps. In particular, *Stopping Cores* and *Saturated Bundle Cancellation* already identify the ten three-handle exclusions and ask for a safe adapter. *Genus Budgets and Native Interfaces* supplies an initial source reduction, but not the later normal-cut adapter with its attached interfaces [6, 7, 8, 9]. Appendix B records the full targeted overlap audit.

The contributions proved or implemented here are:

1. A cap-incidence formula for exceptional normal-cut two-handles and a relative top-handle elimination theorem in the authenticated source charts.
2. A single source tree that specializes to valid traces for every admissible normal coordinate vector, with coordinate-independent trace size.
3. An exact short-row forest criterion and deficiency formula for the abstract joint deletion problem, together with an independent prototype.
4. A homological rigidity theorem showing that much of that apparent forest-choice freedom disappears on genuine geometric inputs.
5. An exact exceptional-row quotient for a fixed forest, which retains arbitrary long rows and gives a finite pattern kernel.
6. A native version-6 certificate adapter, reproducible coverage gains, and a controlled local scheduling benchmark with its common costs exposed.

1.3 Notation and input size

Let $\mathcal{T}$ be a finite triangulation of a connected compact orientable three-manifold M with nonempty boundary. The mathematical source arguments allow several boundary components. The native validator in this release accepts the narrower class with exactly one torus boundary. It checks reciprocal face gluings, orientability, connectedness, sphere or disk vertex links, circle or interval edge links, and absence of edge reversal. It does not prove that M came from a knot diagram.

A normal vector x has seven nonnegative integer coordinates in every tetrahedron, satisfies the matching equations, and uses at most one quadrilateral type per tetrahedron. Its normal surface is denoted S_x. Zero coordinates and the empty surface are allowed. The metric cut $M \setminus\setminus S_x$ is the completion after splitting along the local sheets; this convention also handles one-sided components. No two-sidedness assumption on S_x is needed for the local results.

Symbol	Meaning
t	Number of source tetrahedra
b_x	$\sum_{i=1}^{7t}\lceil\log_2(x_i+2)\rceil$, total coordinate bits
K	Complementary exceptional handle core of the supplied cut
$\mathcal{P}$	Complete labelled interface rectangles of the parallelity pieces
q, m	Numbers of core three-handles and two-handles
E, C	Two-handle row labels and surviving one-handle column labels
G	Cap graph, with a formal exterior root ρ
H	Occurrence graph for rows of capped length at most two
$r_G(Y)$	Graphic rank, $\lvert V(G)\rvert - \mathrm{comp}(G[Y])$

The source triangulation takes $O(t\log(t+2))$ bits. The full input size for the local operation is thus $O(t\log(t+2)+b_x)$, apart from already supplied source certificates. A word-operation estimate on dense source IDs will always be distinguished from a bit-complexity bound. In particular, an $O(t)$ trace is not a claim that reading arbitrarily large coordinates costs $O(t)$ bit operations.

2 Relative elimination of source top handles

2.1 The source handle core

The source-parallelity decomposition divides the metric cut into repeated product pieces and a bounded exceptional core. Repeated normal disks, face strips, and edge caps are recorded as affine interval families. Exceptional tetrahedron chambers give zero-handles; exceptional paired-face regions give one-handles; extreme gaps of interior source edges give two-handles; and interior source vertices give three-handles. All attaching records retain source tetrahedron, face, edge, and rank flags.

For the current constructor the verified inventory bounds are

$$h_0 \le 6t, \qquad h_1 \le 8t, \qquad m = h_2 \le 12t, \qquad q = h_3 \le 4t, \qquad |\mathcal{P}| \le 66t. \tag{1}$$

These bounds concern explicit exceptional objects; the represented parallelity pieces can have much larger multiplicity and topology. Their complete interfaces, not merely their homeomorphism types, are part of the source packet [6].

Choose the canonical orientation of each global source edge e. Its two oriented endpoint occurrences are a_e, b_e, even if they represent the same global vertex. Let $w_e = |S_x \cap e|$. The w_e intersection points divide e into gaps labelled $0, 1, \dots, w_e$. An interior source edge contributes one exceptional two-handle if $w_e = 0$ and two if $w_e > 0$. Interior gaps $1, \dots, w_e - 1$ belong to the selected product region.

For a source vertex v, put $\iota(v) = v$ when v is interior and $\iota(v) = \rho$ when it lies on the original boundary. The symbol ρ also denotes an end on the new normal-cut boundary.

Lemma 2.1 (Exact cap incidences)**.** *For an interior source edge e, the two belt-disk end occurrences of its exceptional two-handles are*

condition	*gap rank*	*cap endpoints*
$w_e = 0$	0	$(\iota(a_e), \iota(b_e))$
$w_e > 0$	0	$(\iota(a_e), \rho)$
$w_e > 0$	w_e	$(\rho, \iota(b_e))$.

The two occurrences are retained with multiplicity. In particular, a zero-weight loop at an interior vertex has two contacts with its three-handle.

Proof. Before taking any quotient of source flags, take a small product neighbourhood of each open edge gap. Transversely it is a disk; longitudinally it is the indicated subinterval of the oriented edge. The transverse disk times this interval is its dual two-handle. An endpoint at a source vertex meets the vertex neighbourhood, which is a three-handle exactly when the vertex link is a sphere. A boundary vertex has a disk link and contributes an exterior end instead.

An endpoint at an intersection with a normal disk lies on a boundary sheet of the metric cut. Thus a zero-weight edge keeps both source-vertex ends, whereas each extreme gap of a positive-weight edge has one source-vertex end and one cut-boundary end. The intervening gaps have two normal-sheet ends and occur in the product families. The source edge-link and orientation checks ensure that this description descends through all face identifications. A global equality $a_e = b_e$ does not identify the two oriented end flags. The argument is local to the split sheets, so a one-sided global normal component causes no exception. □



Figure 1: The cap graph records the extreme gaps. The common drawing of ρ is a scheduling convention; distinct boundary pieces are never identified in the manifold. Boundary source vertices also map to ρ.

2.2 The relative geometric hypothesis

A graph trace alone cannot justify a marked three-manifold operation. The entire protected interface must avoid the support of the cancellation. Even the assertion that ports lie on zero- and one-handles is insufficient: a three-handle attaching sphere can meet patches of those lower handles. The native argument uses the actual rank charts.

Lemma 2.2 (Separation in the source rank charts)**.** *In the source-parallelity decomposition, the labelled interface rectangles $\mathcal{P}$, with compatible corner collars, can be realized disjointly from the interiors and attaching regions of the exceptional two/three-handle cancellation supports. This realization respects all source identifications and the specified port charts.*

Proof. Work first in the split tetrahedra, with their source flags kept distinct. Choose mutually compatible small regular neighbourhoods of vertex sectors, edge-gap sectors, and paired-face sectors. The normal disks bound the split regions and separate their successive gap interiors. This is a finite PL construction for each supplied x; it is used to prove the interpretation of the compressed families, not to enumerate them.

A prism family has ranks $1 \le r < n$ between successive parallel disks of one type. A face-strip family has ranks $1 \le r < a$ between successive parallel normal arcs at its corner. An edge-cap family has ranks $1 \le r < w_e$. These are strict interior ranks. The vertex sectors lie at the source ends of the ordered gaps. The exceptional two-handles occupy rank 0, and also rank w_e when $w_e > 0$. Thus a selected product block does not occupy a source vertex sector or an exceptional end-gap sector.

When an extreme gap and an interior product gap abut the same normal disk before cutting, their closures lie on different local boundary copies in the metric cut. This local separation also applies to a one-sided surface.

The core-facing free sides of a face-strip or edge-cap block are the recorded port rectangles. Their source chamber and face flags are retained, including for self-paired tetrahedra. If closures meet a lower-dimensional corner during the initial subdivision, choose product collars on the two flagged sides and round inside their respective sectors. Strict interior rank places the port on the side separated from the end-gap/vertex neighbourhood; no passage through a normal disk is required. Since the paired rank maps agree along a source face, the collar choices can be made equivariantly under each pairing. Edge and vertex links being a circle, interval, sphere, or disk give precisely the local PL neighbourhoods needed for simultaneous rounding.

After this compatible rounding the full port rectangles and their collars miss the vertex balls and the selected exceptional two-handles, including their lateral attaching annuli and their incident vertex-ball patches. The proof uses the full source product blocks. It would not follow from a list of port owner dimensions or a matrix of signed exponents. □

Remark 2.3. The protected set in lemma 2.2 is the native parallelity interface. A future inherited boundary-pattern curve can cross a vertex patch or an extreme-gap support even when it is carried by a zero/one skeleton. Such a curve requires a separate avoidance or transport proof. This article does not silently enlarge $\mathcal{P}$ to an arbitrary pattern.

2.3 One cancellation and a complete rooted schedule

Lemma 2.4 (Relative single-contact cancellation)**.** *Let h be an exceptional two-handle with exactly one contact, counted with multiplicity, with the currently remaining three-handles. Let that contact be with D. Assume that the other cap of h is on the current exterior boundary and that the marked interface misses the cancellation support. Removing h and D gives a homeomorphic manifold relative to that interface. Each surviving two-handle keeps its original attaching cycle to the zero/one-handle union. Its cap contacts with D become exterior contacts.*

Proof. This is a two/three-handle cancellation with geometric incidence one. In the source handle charts, h and D are balls glued along their unique common cap disk. Their union is a ball. Its attachment to the remaining handles is a disk: the complementary disk in the attaching sphere of D is extended by the lateral attaching annulus of h. The other cap is exterior. Removing a ball attached along a boundary disk preserves the homeomorphism type by adjusting a collar of that disk.

The collar can be chosen away from the protected interface. This gives a homeomorphism fixed on the interface and its specified charts. It does not assert that every point of every undeleted lower handle is fixed: the attaching-disk collar must move. The reduced presentation itself consists of the undeleted handles, whose mutual attaching records remain unchanged. In particular, the other two-handle attaching cycles are literally the same records. Only their upper cap status changes when D is erased. □

Define the cap multigraph G_x on the q interior vertex three-handles and the formal root ρ. Each exceptional two-handle is an edge with the endpoint occurrences in lemma 2.1. An exterior–exterior handle is a loop at ρ. A loop at an interior vertex is also retained as a loop, with two incidences.

Lemma 2.5 (Rooted deletion criterion)**.** *For any such cap graph, all three-handles can be deleted by single-contact steps if and only if every graph vertex reaches ρ. The edge sets of complete schedules are exactly the spanning trees of G_x.*

Proof. Given a spanning tree rooted at ρ, delete each vertex after its parent, using the edge to that parent. At that instant the edge has exactly one live endpoint. Conversely, orient each chosen edge from the vertex it deletes toward its other end. That end is ρ or a previously deleted vertex. Following the chosen edges therefore reaches ρ and never forms a cycle. Exactly q edges are selected on $q+1$ vertices, so they form a spanning tree. A component not meeting ρ cannot supply its first legal step. A loop never has exactly one live endpoint. □

Theorem 2.6 (Source-relative top-handle elimination)**.** *For every admissible normal vector in the stated source setting, all q source three-handles can be removed by exactly q exceptional two/three cancellations. The resulting core K' has no three-handles and satisfies*

$$(K, \mathcal{P}) \cong (K', \mathcal{P}).$$

All labelled port rectangles and charts, all zero/one-handle IDs, and all surviving two-handle attaching cycles can be retained. Reattaching the same parallelity pieces along the same interfaces gives a homeomorphic whole cut.

Proof. Consider a component U of the graph of zero-weight source edges on the interior vertices. A positive-weight edge incident to U gives an exterior contact by lemma 2.1. A zero-weight edge from U to a boundary vertex does the same. If neither occurred, every source edge incident to U would have weight zero and both ends in U. This would isolate U in the source one-skeleton, contrary to connectedness of that skeleton and nonemptiness of the source boundary. Hence every such component reaches an exterior contact, and G_x is rooted-connected.

Use lemma 2.5 for the schedule, lemma 2.2 for its protected interface, and lemma 2.4 for each step. Induction gives the relative homeomorphism. Boundary components represented by the same formal root are never glued together: every actual cancellation has only one live three-handle end. Finally, a homeomorphism fixed on the entire gluing interface extends by the identity over every reattached parallelity piece. □

Deleting equal numbers of two- and three-handles preserves Euler characteristic. Surviving two-handle IDs may be densely renumbered for the old terminal consumer after source replay, but a certificate must record the original IDs used by the trace. This distinction prevents an accidental reassignment of attaching cycles when rows are removed.

3 One source tree for every normal cut

The cap graph changes when a source edge weight becomes positive. It would be wasteful to regard each zero/nonzero support as an unrelated scheduling problem. The source one-skeleton already supplies one schedule template for all of them.

Definition 3.1 (Boundary-rooted source forest)**.** Choose a forest T_0 in the source vertex graph such that every interior vertex has a unique parent edge and every component has one source boundary vertex as its root. There is no parent edge for a boundary vertex. Record the interior vertices in a parent-before-child order. Such a forest is produced by a multi-source breadth-first search starting at all boundary vertices.

The term forest refers to the actual source vertices. Contracting all boundary roots to one formal vertex makes its interior portion a rooted tree for the scheduling argument. This is still only graph bookkeeping.

Theorem 3.2 (Universal source-tree specialization)**.** *Fix T_0 independently of normal coordinates. For every admissible x, process its interior vertices in the recorded order. For a vertex v with parent edge e, choose*

1. *the unique exceptional gap of e, when $w_e = 0$;*
2. *the extreme gap of e adjacent to v, when $w_e > 0$.*

Cancel that two-handle with the three-handle of v. This is a complete valid trace for the relative operation of theorem 2.6. The same source template works for all admissible coordinate supports and all scales.

Proof. An edge incident to an interior vertex is not a boundary edge, so the required exceptional gap exists. If its weight is zero, its other endpoint is the parent, which is a boundary vertex or has already been removed. It therefore has exactly one live three-handle contact, namely v. If its weight is positive, the child-adjacent extreme gap has its other endpoint on the new cut boundary; it is immediately eligible, regardless of the parent's status.

Distinct vertices use distinct parent edges of the source forest. Two different choices can consequently never request the same exceptional two-handle, including after a support change replaces one zero gap by two extreme gaps. A parent edge cannot be an interior loop. Every interior vertex is processed once, and lemma 2.4 applies at every step. The construction makes no assumption on coordinate size and includes $x = 0$. □

Corollary 3.3 (Support and scale invariance)**.** *The cap graph, considered with oriented extreme-gap labels, depends on x only through the predicates $w_e = 0$. Positive scaling $x \mapsto kx$ leaves the source-tree choices and the corresponding dense two-handle IDs unchanged. A specialized trace has the same number of moves and source-ID bits at every positive scale.*

Proof. The cap formula only distinguishes zero from positive weight. In the latter case the numerical label w_e changes but the order of the two extreme gaps does not. Source vertices, source edge order, and the parent forest are fixed. The trace records source handle IDs, while its input fingerprint has fixed encoded length. □

The corollary concerns the added top trace. The complete source certificate still contains exact coordinates and affine endpoints and can grow with b_x. It would be incorrect to infer constant total proof size from constant top-trace size.

3.1 A linear queue and an independent trace checker

The default implementation uses a queue rather than requiring the reusable template. Initialize every edge with one live endpoint. When a vertex is deleted, revisit its incident edges and enqueue newly eligible ones. Duplicate queue entries are harmless: a popped edge is checked against the current live sets before use. Each interior loop is stored once in the vertex adjacency list but retains both endpoint occurrences in the edge record.

Proposition 3.4 (Local cost and trace size)**.** *Given an authenticated dense-ID cap packet with m edges and q interior vertices, a complete rooted schedule and the surviving edge list can be produced in $O(m+q)$ word operations and $O(m+q)$ space. The supplied trace can be checked in the same structural bound. For the source cores in* (1)*, the additional proof takes $O(t\log(t+2))$ bits. Source validation, coordinate reading, rank address reconstruction, and fingerprinting remain polynomial in $t\log(t+2)+b_x$.*

Proof. The initialization and final dense-ID survivor scan visit each edge once. There are at most $2m$ vertex–edge incidences. Each is revisited only when its vertex is deleted. Each visit performs constant work on an edge's two endpoint occurrences and can enqueue it at most once. The initial queue also has at most m entries. Thus all queue pops and endpoint checks have linear total count. A checker reads each of the q pairs, rejects reuse, and tests its at most two live contacts. It finally compares the exact survivor list.

All original handle IDs are $O(t)$, so the q pairs and at most m survivors use $O(t\log(t+2))$ bits. The template also consists of $O(t)$ vertex, parent, and source-edge labels of this length. Reconstructing the source data requires a polynomial number of combinatorial operations and arithmetic on the input integers. No loop ranges over the numerical normal weight, and the fingerprint uses hexadecimal coordinate encodings rather than decimal conversions with implementation-specific digit limits. □

The public template functions validate their source arguments rather than assuming a previously trusted cache. Thus their current complete-call cost includes source validation even when the combinatorial template is reused. A future authenticated session object could remove duplicate validation work while preserving the source binding. Such caching is a separate implementation improvement, not part of the linear queue claim.

3.2 How the source orientation is authenticated

A triangulation edge can appear in many local tetrahedra, and its two global endpoint vertices can coincide. The canonical orientation is determined by the signed source edge union, not by sorting vertex labels. The producer uses the union-find root's own oriented local representative. The verifier uses a different reconstruction: it scans *every* local occurrence, reverses its endpoint pair according to the recorded orientation parity, and demands agreement.

This matters in practice. The general source validator keeps a convenient last-seen endpoint pair that is sufficient for link counting but need not be in the canonical orientation. Using that pair to decide which extreme gap has rank zero can reverse the cap formula. The new proof checker does not trust that shortcut. It reconstructs the complete expected exceptional rank inventory before interpreting any move.

3.3 The exact scope of uniformity

The template is universal for one fixed source triangulation. A subsequent hierarchy state may have a different triangulation or a differently represented marked handle structure. The theorem does not identify those states, transport their patterns, or establish a bounded number of source rebuilds. Its useful implication is more local: within a fixed source, top-handle elimination introduces no branch over normal supports and no normal-weight-dependent number of moves.

4 Joint forest selection in the abstract occurrence model

The top-handle theorem allows many possible deletion trees. The rows left behind can differ, so it is natural to ask whether a tree can be chosen to make the subsequent one/two-handle cancellation easy. This section solves an exact abstract version of that problem. The geometric constraint in the next section is indispensable when interpreting the answer.

4.1 Raw occurrences, saturation, and peeling

Fix the primal forest used to cancel the zero/one-handle forest. The remaining one-handles have labels C, and the original two-handles have labels E. Let

$$a_{ec} \in \mathbb{Z}_{\geq 0}$$

be the actual unsigned number of passages of the attaching curve of e through the cocore of c. These are geometric occurrences, not freely reduced words or signed exponent sums. For the native core they are obtained by counting the dimension-one flags of the source attaching cycles after omitting the primal forest columns.

Definition 4.1 (Unit-column peeling)**.** For a live row set $R \subseteq E$, a pivot (e, c) is a unit-column pivot if $e \in R$ and

$$\sum_{f \in R} a_{fc} = a_{ec} = 1.$$

Peeling this pivot deletes row e. The other rows remain unchanged, and column c becomes zero. A row set is peelable if all its rows can be deleted this way.

Put $b_{ec} = \min(2, a_{ec})$. Every unit-column predicate on every row subset is unchanged by this entrywise saturation: a nonnegative sum is one exactly when it consists of one unit contribution and otherwise zeros. We therefore use b for peeling, retaining a when integral or mod-two chain information is needed.

A nonempty row set Z is a *stopping set* if each of its column sums is zero or at least two. Its first row can never be peeled, even with other rows present. Conversely, a stalled peeling process leaves a stopping set. Unions of stopping sets are stopping sets, so their largest union is the canonical residual. These statements and the equivalence with complete positive raw cancellation are prior results in the incoming *Stopping Cores* report [7]. The present paper uses them within that exact positive model. Allowing geometric slides with intersection cancellation or changing the cocore system is a different problem.

No arbitrary columns may be discarded. If private bundle columns are omitted, an independent geometric projection theorem must already justify the omission, as in the saturated-bundle work. The combinatorial theorems below can be used on any designated matrix, but a manifold interpretation requires an authenticated producer for precisely that matrix.

4.2 Short rows form a graph

Call e short if $\sum_c b_{ec} \leq 2$. Introduce an auxiliary ground vertex ω, distinct from the cap-graph root ρ. The occurrence multigraph H has vertices $C \cup \{\omega\}$ and one labelled edge for every short row:

Nonzero entries of the short row	Its edge in H
One unit at c	$c\omega$
Two units at distinct c, d	cd
One entry at least two at c	A loop at c
No entries	A loop at ω

The last case is necessary: an empty two-handle row cannot be deleted by a one/two-handle pivot. Parallel edges and loops are genuine cycles.

Lemma 4.2 (Graphic peeling criterion)**.** *A set of short rows R is completely peelable if and only if $H[R]$ is a forest.*

Proof. At a nonground vertex, the graph degree equals the saturated column sum, with a loop counted twice. A nonempty forest component has at least two leaves. At most one leaf is ω, so there is a nonground degree-one vertex and therefore a legal unit-column pivot. Removing its edge preserves the forest, and induction peels every row.

Conversely, the edges of any cycle form a stopping set. At every nonground vertex their degree is zero or at least two. This includes an interior loop, a loop at ω, and a two-edge cycle of parallel edges. None of these edge sets can have a first peeled row. □

4.3 The two-forest criterion

First suppose every row is short, and let G be the connected cap graph on the same edge label set E. By lemma 2.5, a complete top schedule deletes a spanning tree F of G. Its complement must be an H-forest if the residual rows are to peel.

Theorem 4.3 (Abstract joint forest criterion)**.** *For connected G and short-row graph H, the following are equivalent:*

1. *There is a spanning tree F of G such that $H[E \setminus F]$ is a forest.*
2. *E is a disjoint union of a G-forest and an H-forest.*
3. *For every $Y \subseteq E$,*
$$|Y| \le r_G(Y) + r_H(Y). \tag{2}$$

A deterministic polynomial-time procedure finds the two schedules or a subset violating (2).

Proof. The first condition immediately gives the second. Given a partition into forests I_G, I_H, extend I_G to a spanning tree by moving edges from I_H. If I_G does not yet span, connectedness of G supplies an edge of I_H between different I_G components. Adding it preserves the G-forest; removing it preserves the H-forest. Repeat. This proves the first condition and gives the two schedules by lemmas 2.5 and 4.2.

Intersecting a forest partition with Y proves necessity of (2). Sufficiency is the two-matroid partition theorem of Edmonds–Fulkerson [4, Theorem 1c]. For completeness, the explicit augmenting proof and its witness are given below. □

Now let
$$D = \{e : \textstyle\sum_c b_{ec} > 2\}, \qquad E_0 = E \setminus D.$$

Consider the restricted strategy that requires all rows of D to be deleted by the top-handle forest. This is a restriction of the strategy, not a necessary condition for raw cancellation. The single row $(1, 1, 1)$ is already peelable.

Corollary 4.4 (Forced long rows)**.** *The restricted strategy is possible exactly when G is connected, $G[D]$ is a forest, and for all $Y \subseteq E_0$,*

$$|Y| \le r_G(Y \cup D) - |D| + r_H(Y). \tag{3}$$

Here H is the short-row graph on E_0. When D is independent, the first two terms on the right are $r_{G/D}(Y)$.

Proof. A spanning tree can contain D only when D is a forest. Contracting this forest identifies spanning trees containing D with spanning trees of G/D. Graphic matroid contraction has rank $r_{G/D}(Y) = r_G(Y \cup D) - r_G(D)$, and $r_G(D) = |D|$. Apply theorem 4.3. Contraction can create loops and parallel edges; they remain in the instance. □

4.4 A constructive proof with a failure certificate

Write $G' = G/D$, with $D = \varnothing$ in the unforced case. Maintain disjoint sets $I_0, I_1 \subseteq E_0$, independent in G' and H, respectively. All other edges are unassigned. An edge $e \notin I_i$ is an i-sink if adding it to I_i preserves a forest. Otherwise its endpoints are joined by the unique path in the forest I_i. Draw an exchange arc $e \to f$ for each path edge f. For a loop this path has no edges and there is no exchange arc.

Run breadth-first search in this exchange digraph, starting at every unassigned edge. If a shortest source-to-sink path $e_0, e_1, \dots, e_\ell$ is found, put its terminal edge into its sink forest. For every $j < \ell$, assign e_j to the old forest of e_{j+1}. The number of assigned edges increases by one.

Here is an explicit independence argument. Represent graph edges by their endpoint incidence vectors over $\mathbb{F}_2$. Forest edge vectors are independent. For an incoming edge e_j, its expansion in its target forest has coefficient one on e_{j+1}. It has coefficient zero on every later removed edge e_{k+1} of that same forest, $k > j$: otherwise there would be an exchange arc skipping part of the shortest path. The matrix of these coefficients on the removed edges is triangular with diagonal one. Replacing the removed forest edges by the corresponding incoming edges therefore preserves the old span and independence. No earlier path vertex was a sink, since that would give a shorter path. The terminal edge lies outside the old span of its sink forest and increases its rank by one. This proves the augmentation rule.

If no sink is reachable, let Y be the reached edge set and U the unassigned set. Then $U \subseteq Y$. Every fundamental path reached in either forest lies entirely in Y. Consequently

$$r_i(Y) = |I_i \cap Y| \quad (i = 0, 1), \qquad |Y| - r_0(Y) - r_1(Y) = |U|.$$

When $U \neq \varnothing$ this is a violated rank inequality. It also proves optimality of the current partial partition: any two-forest union has size at most

$$|E_0 \setminus Y| + r_0(Y) + r_1(Y) = |E_0| - |U|,$$

and the current partition attains this value. When $U = \varnothing$, all edges are assigned and the desired full partition has been found.

For $m_0 = |E_0|$ and $v = |V(G')|+|V(H)|$, the simple implementation has at most m_0 augmentations. Each search visits at most m_0 edge labels and uses ordinary forest-path traversals. Its conservative structural bound is

$$O\big(m_0^2(v + m_0) + m_0^3 \log(m_0 + 2)\big).$$

Integer parsing, label comparisons, and source authentication are additional polynomial bit costs. Large occurrence values are compared with two, never expanded into repeated crossings by this planner.

4.5 An exact residual-deficiency theorem

When the restricted strategy cannot finish, its failure need not be reported only as a boolean. Assume G is connected and D is a forest. Define

$$d = \max_{Y \subseteq E_0} \big(|Y| - r_{G/D}(Y) - r_H(Y)\big). \tag{4}$$

The empty set makes $d \geq 0$.

Theorem 4.5 (Optimal residual nullity)**.** *The deficiency satisfies*

$$d = \min_{\substack{F \text{ spanning tree of } G \\ D \subseteq F}} \big(|E \setminus F| - r_H(E \setminus F)\big). \tag{5}$$

The exchange procedure produces an attaining tree, an occurrence forest in its residue, and a rank witness certifying the common optimum.

Proof. The augmentation argument proves that the maximum two-forest union in E_0 has size $m_0 - d$. For any tree $F \supseteq D$, let J be a maximum H-forest in its residue $R = E \setminus F$. Then $F \setminus D$ and J are a two-forest union, so

$$|F \setminus D| + r_H(R) \leq m_0 - d.$$

Since $|F \setminus D| + |R| = m_0$, the residual nullity is at least d.

For the reverse inequality, take a maximum partial partition I_G, I_H. Every unassigned edge has endpoints in one I_G component; otherwise it could be assigned to that forest. Hence $I_G \cup I_H$ spans the connected graph G/D. Extend I_G to a spanning tree T using edges of I_H only. The unused part $J \subseteq I_H$ remains an H-forest. Its complement in $E_0 \setminus T$ consists of exactly the d unassigned edges. Therefore

$$|E_0 \setminus T| - r_H(E_0 \setminus T) \leq |E_0 \setminus T| - |J| = d.$$

Lift T through the contracted forest D. This attains the lower bound. □

The number d is the minimum number of *additional arbitrary row deletions* that would make an optimized residue a forest. Such deletions are not licensed geometric cancellations and are never performed by the prototype. The certificate measures the exact abstract obstacle; it is not a proof that the manifold is not a handlebody.

4.6 Two instructive limitations

Example 4.6 (An abstract exchange gain with no geometric realization)**.** Take two interior cap vertices u, v and three rows:

e	cap edge in G	single-column occurrence row
0	uv	(1)
1	$v\rho$	(0)
2	$u\rho$	(2).

Deleting edges $0, 1$ leaves the unpeelable row (2); deleting $1, 2$ leaves (1) and succeeds. This is a genuine test of the abstract exchange algorithm. It is not an example of a three-manifold improvement. Over $\mathbb{F}_2$, its occurrence boundary and cap incidence have nonzero product: each interior vertex column sees the unit contribution from row 0. It violates $\partial_2\partial_3 = 0$. Section 5 explains the consequence of imposing that identity.

Proposition 4.7 (General peelability is not a matroid)**.** *The peelable row subsets of an arbitrary nonnegative occurrence matrix need not satisfy the matroid augmentation axiom.*

Proof. Use the four binary rows

$$a = (0, 0, 1), \quad b = (0, 1, 1), \quad c = (1, 1, 0), \quad f = (1, 1, 1).$$

Both $I = \{c, f\}$ and $J = \{a, b, c\}$ peel: in I first use the third column of f; in J first use the first column of c, then the second of b, then the third of a. Yet $|J| > |I|$ and neither $I \cup \{a\}$ nor $I \cup \{b\}$ peels. Every nonzero column sum in either enlarged set is at least two. Thus augmentation fails. □

This counterexample does not prove hardness of the general geometric problem. It identifies the precise reason that the short-row matroid argument cannot simply be extended to arbitrary long rows.

5 The chain constraint: when forest optimization disappears

The two-graph model has an additional constraint on native geometric inputs. Its cap incidences and raw attaching counts are parts of the same cellular chain complex. This constraint makes a substantial part of the apparent forest-choice problem collapse to one tree extension. It also provides a necessary realizability check for proposed abstract examples.

5.1 Raw parity and complementary tree coordinates

Fix an authenticated core K and a geometric forest contraction among its zero- and one-handles. Let E index its m two-handles, let C index the remaining one-handles, and let q be the number of three-handles. Write a_{ec} for the *uncapped* geometric occurrence count of one-handle c in the attaching cycle of two-handle e. Define

$$A = (a_{ec} \bmod 2) \in \mathbb{F}_2^{E\times C}, \qquad B \in \mathbb{F}_2^{E\times\{1,\dots,q\}},$$

where B_{ev} counts the cap occurrences of e at three-handle v modulo two. Both ends of a loop are counted. Orientations disappear over $\mathbb{F}_2$, but geometric occurrences must be counted before reduction modulo two.

The handle chain complex gives

$$\partial_2 = A^{\mathsf{T}}, \qquad \partial_3 = B, \qquad A^{\mathsf{T}}B = 0. \tag{6}$$

The matrix B^{T} is the reduced incidence matrix of the cap graph G, with the formal exterior root omitted. By theorem 2.6, G is connected in the native source setting. Hence $\operatorname{rank} B = q$, and

$$\beta_2(K;\mathbb{F}_2) := \dim_{\mathbb{F}_2} H_2(K;\mathbb{F}_2) = m - \operatorname{rank} A - q. \tag{7}$$

Several core components cause no problem when each reaches an exterior end: the common graph root is formal and the three-handle columns remain independent. Closed components with nonzero H_3 require a different rank statement.

Suppose first that every *raw* row has length at most two. Construct the occurrence graph H as in section 4: singleton rows join their column to a ground vertex, two distinct occurrences join their columns, and double occurrences or empty rows are loops. Then A^{T} is exactly a reduced incidence representation of H. Write $r_H(S)$ for its graphic rank on edge set S, and $\nu_H(S) = |S| - r_H(S)$ for its cycle nullity.

Theorem 5.1 (Geometric short-row invariance)**.** *If $\sum_c a_{ec} \le 2$ for every $e \in E$, then every spanning tree F of the cap graph satisfies*

$$r_H(E \setminus F) = r_H(E), \qquad \nu_H(E \setminus F) = \beta_2(K;\mathbb{F}_2). \tag{8}$$

Consequently, one such residue admits complete unit-column peeling if and only if every such residue does. Changing the top-handle tree cannot improve its occurrence-graph nullity.

Proof. Transposing equation (6) gives

$$\operatorname{col}(A) \subseteq \ker(B^{\mathsf{T}}),$$

the cycle space of G. Projection onto the coordinates $E \setminus F$ is injective on this cycle space: a vector in its kernel would be a cycle supported entirely on a forest, and therefore zero. The same projection is injective on $\operatorname{col}(A)$, so $\operatorname{rank} A_{E\setminus F} = \operatorname{rank} A$. These ranks are respectively $r_H(E \setminus F)$ and $r_H(E)$.

Every cap spanning tree has q edges. Substitution in equation (7) proves the nullity identity. Finally, the short-row peeling criterion of section 4 identifies complete peeling with zero occurrence-graph nullity. □

This is a restriction on the optimization problem, rather than a defect in the abstract matroid theorem. In compatible geometric inputs, the row space representing H lies in the cycle space of G. The two graphs cannot be varied independently.

5.2 Mandatory rows and the homological criterion

Capping counts at two preserves singleton predicates but can change parity. For example, a raw row (3) has binary row (1), whereas its capped row (2) represents a loop and hence a zero incidence column. With no three-handles, this example has $m - \operatorname{rank} A - q = 0$ but occurrence-graph nullity one. Thus theorem 5.1 cannot simply be applied to arbitrary capped short rows.

There is an exact repair for the restricted strategy. Put

$$\begin{aligned} D_{\text{long}} &= \{e : \sum_c \min(2, a_{ec}) > 2\}, \\ D_{\text{loop}} &= \{e : a_e = 0 \text{ or } a_e = k\mathbf{e}_c \text{ for some } k \geq 2\}, \\ \overline{D} &= D_{\text{long}} \cup D_{\text{loop}}. \end{aligned} \tag{9}$$

The first set is required to be removed by the restricted rule in section 4. The second consists of all occurrence-graph loops, none of which can remain in a completely peelable residue. Thus every successful certificate in that strategy must place all of $\overline{D}$ in its cap tree. Every remaining row is genuinely one unit occurrence or two distinct unit occurrences; its raw and capped parities agree.

Theorem 5.2 (Mandatory-forest reduction)**.** *Assume $\overline{D}$ is a forest of G. Set $k = |\overline{D}|$, $E_1 = E \setminus \overline{D}$, and let H_1 be the occurrence graph on E_1. For every cap spanning tree $F \supseteq \overline{D}$,*

$$\nu_{H_1}(E \setminus F) = (m - k) - (q - k) - r_{H_1}(E_1) = \beta_2(K; \mathbb{F}_2). \tag{10}$$

For the fixed primal forest and the restricted short-row rule, a complete top-deletion and raw-peeling certificate exists exactly when $\overline{D}$ is a cap forest and $\beta_2(K; \mathbb{F}_2) = 0$.

Proof. Let Q have one column for each component of $G[\overline{D}]$ not containing the exterior root. Its entries are one on the three-handle vertices in that component and zero elsewhere. Thus

$$B_{\overline{D}} Q = 0, \qquad B_1 := B_{E_1} Q$$

is the transpose of the reduced incidence matrix of $G/\overline{D}$. Multiplying equation (6) by Q gives

$$A_{E_1}^{\mathsf{T}} B_1 = 0. \tag{11}$$

Because $\overline{D}$ is a forest, the contracted connected graph has $q - k + 1$ vertices. The projection argument in theorem 5.1 applies to A_{E_1}, B_1. It shows that restriction away from the tree $F \setminus \overline{D}$ preserves $r_{H_1}(E_1)$, proving the first equality in equation (10).

To identify this value with the original core's homology, use that $B_{\overline{D}}$ has full row rank k. Choose a right inverse R with $B_{\overline{D}} R = I_k$. The original chain identity implies

$$A_{\overline{D}}^{\mathsf{T}} = A_{E_1}^{\mathsf{T}} B_{E_1} R.$$

The deleted column maps of ∂_2 are therefore combinations of the surviving ones. Consequently

$$\operatorname{rank} A = \operatorname{rank} A_{E_1} = r_{H_1}(E_1).$$

Equation (7) now proves the second equality.

Every successful restricted certificate has a cap tree containing $\overline{D}$, so this set must be a forest. Its residue has nullity zero, forcing $\beta_2 = 0$. Conversely, if both conditions hold, extend $\overline{D}$ to any cap spanning tree. Its residue is an occurrence forest, and therefore peels completely. The rooted schedule of lemma 2.5 and the authenticated geometric cancellation rules turn these graph steps into a core cancellation certificate. □

The contraction in this proof chooses an edge set; it does not assert that the mandatory edges alone form an initial legal cancellation sequence. After extension to F, the complete geometric trace is ordered from the exterior root as in lemma 2.5.

Corollary 5.3 (One extension suffices)**.** *On supplied authenticated incidence data, the restricted strategy can be decided by testing whether $\overline{D}$ is a cap forest, extending it once to a spanning tree when possible, and testing or peeling the residual occurrence graph. No search over alternative extensions is required.*

If the incidence lists are indexed and isolated occurrence vertices are suppressed, let N be the number of nonzero raw incidence entries. Finding $\overline{D}$, traversing its components, building the quotient, choosing a spanning tree and peeling use $O(m + q + N + 1)$ graph work and memory. Component traversals suffice; a general binary-matrix rank computation is unnecessary. Source authentication, index construction and the bit costs of reading or comparing binary counts are separate from this graph bound. The positive output remains an ordinary replayable cancellation trace.

This is not a generic handlebody test based on homology. Sufficiency uses the authenticated handle charts, the fixed primal forest, the mandatory forest hypothesis and the particular short-row rule. A vanishing second Betti number without those hypotheses does not supply the cancellations. Moreover, failure of the restricted strategy can occur for a handlebody. The later one/two cancellations also need their own marking analysis if a consumer requires arbitrary boundary-pattern transport.

5.3 Consequences for abstract evidence and deficiency

The retained abstract exchange regression makes the distinction concrete. Its cap edges are $e_0 = v_0v_1$, $e_1 = v_1\rho$, $e_2 = v_0\rho$, and its raw rows on one occurrence column are $(1), (0), (2)$. The cap tree $\{e_0, e_1\}$ leaves an occurrence loop, whereas $\{e_1, e_2\}$ leaves a peelable singleton. However, its binary matrices satisfy

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad A^{\mathsf{T}}B = (1, 1) \neq 0.$$

Thus these exact data cannot be the incidences of an authenticated handle core. They remain a useful test of the abstract exchange algorithm, but provide no example of a native geometric speedup.

Example 5.4 (A geometric family with several top trees)**.** Start with a standard solid torus V, presented by one zero-handle and one one-handle c. Attach $r \geq 2$ two-handles along disjoint parallel primitive *longitudes* of V, each crossing the cocore of c once. These curves represent a generator of $\pi_1(V)$; they are not meridians of V.

For an explicit embedding, use the standard genus-one splitting $S^3 = V \cup W$, in which a meridian of the complementary solid torus W is a longitude of V. Take the two-handles to be small neighbourhoods of r disjoint meridian disks in W. Their complement in W consists of r balls. The resulting core is therefore S^3 with r open balls removed, equivalently a ball with $r - 1$ interior balls removed. Fill any $0 \leq q \leq r - 1$ of its spherical boundary components with three-handles.

The r complementary chambers lie cyclically around W. Consequently the cap graph is the r-cycle with the unfilled chamber vertices identified with ρ; for $r = 2$ the cycle has two parallel edges. The occurrence graph has r parallel edges between c and its ground, and rank one. Every cap tree deletes q rows and hence leaves

$$|E \setminus F| = r - q, \qquad \nu_H(E \setminus F) = r - q - 1 = \beta_2(K; \mathbb{F}_2).$$

When $q = r - 1$, there are r different cap trees and every one succeeds; the filled core is a ball. When $r = 3$ and $q = 1$, there are two cap trees, and each leaves nullity one. Thus genuine geometric inputs can have several tree choices and positive invariant defect. The ports in this family are empty; the construction supplies geometric examples only, without a native triangulation benchmark claim.

The exact abstract deficiency of section 4 remains meaningful. For any cap spanning tree F, the rows B_F form an invertible square matrix. Equation (6) then expresses A_F^{T} in terms of $A_{E\setminus F}^{\mathsf{T}}$, so every such restriction retains the raw binary rank $\operatorname{rank} A$. When only D_{long} is forced, passing from the remaining raw short rows to their capped occurrence graph merely zeros the columns arising from odd single multiplicities at least three. It cannot increase rank. Every corresponding occurrence nullity is therefore at least $\beta_2(K;\mathbb{F}_2)$.

If $\overline{D}$ is a forest, theorem 5.2 supplies an extension attaining this lower bound. In that case the abstract minimum deficiency equals the second Betti number as well. If $\overline{D}$ is cyclic, a complete short-row certificate is impossible; optimizing a partial residue can still be a diagnostic. Further native progress must then use information outside this complete short-row forest strategy, such as the structure of retained longer rows, a different primal contraction, or authenticated geometric moves that change the occurrence data.

6 An exact kernel for exceptional rows after a fixed forest

The forced-long-row strategy excludes instances that already admit an easy raw cancellation. A complementary approach keeps the long rows and contracts the short-row structure around them. The resulting theorem concerns a *fixed* choice of top-handle forest and a fixed primal forest. It does not optimize those choices. Its advantage is an exact description of both the remaining decision problem and the source stopping obstruction.

6.1 The quotient and its ground convention

Let $A = (a_{ec})$ be the nonnegative occurrence matrix of the rows retained after the chosen top forest; here A again denotes integer counts, before any reduction modulo two. As in section 4, put $b_{ec} = \min(2, a_{ec})$. Partition its row labels into $S \sqcup L$, where every row of S is short. Usually L consists of all rows with $\sum_c b_{ec} > 2$, but the theorem allows additional exceptional rows. Write $k = |L|$ and form the graph $H[S]$ of lemma 4.2, with column vertices C and ground ω.

If $H[S]$ has a cycle, its edge rows already constitute a stopping set of A. Consequently full raw cancellation is impossible in the designated positive model, irrespective of the exceptional rows. We may therefore assume that $H[S]$ is a forest. In particular its edges have actual unit incidences; a repeated crossing or zero short row would have been a loop. Include isolated column vertices when forming its components. Denote the component containing ω by K_ω, and the set of all other, *free* components by $\mathcal{K}$.

The exceptional-row quotient has row set L, column set $\mathcal{K}$, and entries

$$q_{eK} = \min\left(2, \sum_{c \in K} b_{ec}\right), \qquad e \in L, \quad K \in \mathcal{K}. \tag{12}$$

There is no column for K_ω. Saturating before aggregation gives the same result as aggregating the original counts and then saturating. The sums in (12) count occurrences, not merely the number of incidence sites. Two unit occurrences at different vertices of one tree contribute two.

Example 6.1 (Why the grounded component is omitted)**.** Take

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \qquad S = \{\text{first row}\}, \qquad L = \{\text{second row}\}.$$

The first column belongs to the grounded component, so $Q = (2)$. Both source rows form a stopping set. If the grounded component were retained as a quotient column, the erroneous quotient (1 2) would have a unit pivot. There is no corresponding source pivot: the ground is not an available one-handle, and its incident short edge cannot be removed through the occupied column.

6.2 All stopping projections and their maximal lifts

Allow the empty stopping set. For $R \subseteq L$, mark every column vertex c with $\sum_{e \in R} b_{ec} > 0$, and also mark ω. In each component K of $H[S]$, let $T_K(R)$ be the minimal subtree spanning the marked vertices in K. Equivalently, take the union of all paths between marked vertices. Its edge set is empty when there are fewer than two marked vertices. Let

$$T(R) = \bigcup_K E\big(T_K(R)\big) \subseteq S,$$

using the original short-row labels on these edges. Thus the construction records source rows, not only an isomorphism type of a smaller matrix.

Theorem 6.2 (Exact exceptional-row kernel and obstruction lift)**.** *Assume $H[S]$ is a forest and define Q by (12).*

1. *A subset $R \subseteq L$ is the exceptional-row projection of a stopping set of A if and only if R is a stopping set of Q.*
2. *If R is a stopping set of Q, then $R \cup T(R)$ is the unique largest stopping set of A whose intersection with L is exactly R.*
3. *If $R_* = \mathrm{Stop}(Q)$ is the canonical largest stopping set, then*

$$\mathrm{Stop}(A) = R_* \cup T(R_*). \tag{13}$$

In particular, A is completely peelable if and only if Q is completely peelable. By the prior stopping-persistence theorem, the same equivalence holds for existence of full positive raw cancellation.

Proof. Let P be a source stopping set and put $R = P \cap L$. Suppose a free component K has total exceptional incidence exactly one on R. This incidence occurs at one vertex v, with coefficient one. Consider the subforest consisting of the selected short edges $P \cap S$ in K. If v is isolated in it, the source column sum at v is one, a contradiction. Otherwise its selected tree has at least two leaves. At most one is v, and none is ground. A different leaf has one selected short incidence and no exceptional incidence, again a contradiction. Thus no free component has total exceptional incidence one. Saturation preserves this predicate, so R is a stopping set of Q.

We also claim $P \cap S \subseteq T(R)$. Each nonground leaf of a nontrivial selected short tree must have a positive incidence from R; otherwise its source column sum would be one. Every leaf is therefore marked, with ground allowed as an additional marked leaf. Every selected edge lies on a path between two leaves of its selected tree. That path is the unique path between those vertices in the

original forest, so the edge belongs to $T(R)$. This proves the claim and the maximality assertion once the proposed lift is shown to be stopping.

Conversely, assume R is stopping for Q. At a nonground vertex of $T(R)$ with degree at least two, the selected short rows alone contribute at least two. A nonground leaf is marked and therefore receives at least one exceptional incidence as well as its one short incidence. Its total is again at least two. A vertex with degree zero and positive exceptional incidence can only be the single marked vertex of a free component: otherwise a path to another marked vertex, including ground when present, would meet it. In this case the entire exceptional incidence of that free component is concentrated there. The stopping condition for Q makes that total at least two. All other degree-zero column vertices receive zero. Ground itself is not a matrix column. Hence every source column sum of $R \cup T(R)$ is zero or at least two, proving that it is stopping.

The two directions establish the first two assertions, including empty sets. For the third, every source stopping set projects into R_*, and the marked subtree construction is monotone under inclusion of R. Thus every source stopping set is contained in $R_* \cup T(R_*)$. This set is itself stopping by the second assertion, proving (13). A matrix peels completely exactly when its largest stopping set is empty. Finally, the equivalence between full peeling and full positive raw cancellation is the prior result of [7]; no geometric operation beyond that positive model is introduced here. □

The theorem describes a whole family of obstructions. A quotient stopping set can have several source stopping extensions, but its largest extension has the explicit minimal-subtree formula. The quotient therefore preserves more information than a successful-cancellation boolean or a rank test.

6.3 Constructive schedules and source replay

The decision equivalence has a direct constructive lift. Suppose a quotient pivot deletes exceptional row e at free component K. Among the currently active exceptional rows, the total incidence in K is exactly one. There is therefore a unique source incidence (e, c) in that component, and its coefficient is one. Root the current short tree at c and remove its other leaves inward. Each removed nonground leaf has one short edge and no active exceptional incidence, so its source column is a legal unit leaf. After these removals, (e, c) is itself a legal source pivot. Once this quotient column is used, its unique active exceptional incidence is deleted and no future active exceptional incidence remains in that free component. Pruning it therefore removes no edge needed by the final marked subtree $T(R_*)$.

Apply this procedure to each quotient pivot. If every exceptional row is removed, the remaining short forest peels by lemma 4.2. If the quotient stops at R_*, mark its incidence sites and ground, then remove every unmarked nonground leaf of the remaining short forest. Exactly the minimal subtrees in (13) remain. This gives a maximal source peeling prefix and its exact canonical residual even in an incomplete case.

The lifted trace consists only of pairs of original row and column labels. A verifier can check each pair by summing its input column over the then-active source rows, without trusting the quotient construction. Checking that the final residual has no column sum one certifies maximality. Unlike a producer using general positive pivots, this no-fill trace retains the original entries throughout. General positive pivots may change the labels and entries of a surviving obstruction; their residual row list need not equal $\mathrm{Stop}(A)$.

6.4 A bounded number of exceptional column patterns

The quotient can still contain many free components with identical effects on its k rows. Order L and associate to a quotient column K its signature

$$\sigma(K) = (q_{eK})_{e\in L} \in \{0,1,2\}^k.$$

Delete all-zero columns and retain one representative of each nonzero signature. Denote the resulting matrix by $\widehat{Q}$. Keep every row, including zero rows, and retain the component-to-representative map for lifting to the source.

Corollary 6.3 (Column-pattern kernel)**.** *The matrices Q and $\widehat{Q}$ have exactly the same stopping row subsets. Therefore $\widehat{Q}$ determines* $\mathrm{Stop}(A)$ *through theorem 6.2. It has k rows and at most*

$$\min(|\mathcal{K}|, 3^k - 1)$$

columns, with entries in $\{0,1,2\}$.

Proof. Equal column signatures have equal sums on every subset of the row set L, and hence have the same predicate of being a unit column on that subset. Repeating this predicate adds no stopping constraint. A zero column imposes none. These observations prove preservation of the entire stopping family. There are $3^k - 1$ possible nonzero ternary signatures. □

Preserving zero rows is essential. A row whose incidences all lie in the grounded component becomes a zero quotient row and is a singleton quotient stopping set. Its source obstruction is the row together with the appropriate paths to ground. Removing that row would change the answer. By contrast, the single long row $(1,1,1)$ with no short rows gives three identical unit quotient columns and compresses to (1); its successful pivot is preserved.

6.5 Complexity and the remaining scope

Let N count the explicit row labels, column labels, and nonzero entries, and put $s = |\mathcal{K}|$. Forest testing, component labeling, incidence aggregation, and leaf pruning admit $O(N)$ graph operations. Comparing a binary input count with two requires no expansion in its magnitude. For a simple deterministic implementation, forming dense signatures takes $O(ks)$ ternary entries and lexicographically sorting them takes $O(ks\log(s+2))$ symbol comparisons. Only signatures of components actually present are formed: there is no loop over 3^k possible patterns. Sparse alternatives are available, but unnecessary for the polynomial preprocessing claim. Label sorting and integer input handling contribute their ordinary bit costs. The supplied research implementation also uses direct source scans during lifting; these are polynomial and are not presented as an optimal implementation bound.

The bounded decision matrix and the source lift data serve different purposes. The matrix has at most $k(3^k - 1)$ ternary entries; the forest and the maps needed to reproduce original source rows can retain linear-sized input information. No bound on k in terms of crossing number, hierarchy depth, or coordinate bit length is proved here. Nor does the theorem control how the quotient changes under a different top or primal forest. In particular, proposition 4.7 still prevents replacing the arbitrary-row joint problem by the short-row matroid argument.

For one explicitly supplied matrix, ordinary sparse column peeling already gives a linear-time decision. The kernel is a structural reduction that isolates the exceptional incidence data and

reconstructs exact source obstructions; it is not by itself a faster one-shot recognition algorithm. Its potential use in repeated geometric choices requires further theorems bounding the exceptional interface and the cost of updating it. It also does not supply the cyclic order or boundary-mark transport needed to turn counts into a marked geometric certificate. The construction is new to the present repository audit relative to the inspected incoming packages; we make no assertion of priority over the full literature on stopping sets or graph reduction.

7 Native implementation and certificate composition

7.1 The actual integration point

The change is an adapter inside `normal_cut_handlebodies`, operating after construction of the original compressed parallelity geometry. The marked boundary data are subsequently constructed from the original source packet, independently of this reduction. If the source core has three-handles, the adapter reconstructs its cap packet, runs the rooted queue, and passes the reduced core to the existing terminal paths. These paths include raw core cancellation, disk-bundle absorption, primitive annulus handling, and the finite bundle-boundary kernel. Sources without three-handles follow the existing path and retain their old certificate versions.

Only two previously maintained Python files are modified:

- `fastunknot/normal_cut_handlebodies.py`;
- `fastunknot/normal_cut_handlebodies_verify.py`.

The new source packet, queue, independent top replay, and template API are in three additional modules. The abstract forest-exchange producer and its independent verifier are separate research modules. They do not change the default terminal policy. The exceptional-row and chain experiments are also supplied as research code rather than silently changing recognition dispatch.

The public whole-cut operation accepts a *supplied* normal vector. It does not search for the surface. The resulting histogram says that the components of that particular cut are handlebodies of specified genera and multiplicities. A cut of a nontrivial knot exterior can have such a histogram; this result alone is not an unknot certificate. The new adapter does not change the general exponential guarantee of the complete diagram recognizer.

7.2 Source-bound top proof

The added trace has the following strict schema:

```
{ schema:  'normal-core-top-reduction-v1',
  input_sha256:  source-and-coordinate fingerprint,
  moves:  [[original_two_id, original_three_id], ...],
  surviving_two_handles:  [ascending original IDs] }
```

The fingerprint indexes the exact normalized input. Mathematical replay does not rest on a collision-resistance assumption: the checker also reconstructs the source geometry, exact coordinate rows, full exceptional rank inventory, and endpoint occurrences against which the moves are tested.

The independent top checker imports no producer or queue function. It reconstructs canonical endpoint orientation from all local source edges, checks each proposed pair against the current live

endpoint multiset, rejects repeated or nonexistent IDs, and compares the exact final survivor list. It requires exactly one deletion per source interior vertex. A loop has two contacts and cannot be forged into an eligible singleton by deduplicating its endpoints.

The reduced geometry is constructed only after a successful trace. Every surviving two-handle record is copied with a new dense ID; its source edge, rank, and full attaching cycle stay unchanged. The zero/one handles, parallelity polygons, rank maps, free sides, and ports retain their original records. The original source summary remains a summary of the *original* source decomposition. Its core handle counts should not be misread as post-reduction counts; its Euler characteristic remains valid because equal numbers of two- and three-handles have been removed.

7.3 The whole-cut version-6 wrapper

The enclosing proof stores

```
{ schema:  'normal-cut-handlebodies-v6',
  parallelity:  original source parallelity proof,
  boundary:  original marked boundary proof,
  top_reduction:  the source-bound top trace,
  terminal:  one existing terminal schema v1 through v5,
  census:  the final handlebody histogram }.
```

The nested terminal object has no duplicate source proofs or census and cannot recursively be version 6. The checker validates the outer fields, replays both original source proofs, decodes their exact geometry, applies the independent top trace, and then runs the old terminal verifier on the reduced geometry. This order is essential. Modifying the geometry first and passing the altered object to an original source verifier would discard the source authority needed for the relative argument.

The original six orbit-proof operation counts continue to determine the existing `max_operations` gate. The new top trace has a separate linear source bound and uses the same cooperative interruption callback. It does not consume an undocumented normal-surface enumeration budget. Callback exceptions propagate; a callable with false boolean value is still called. Huge integers use the existing exact hexadecimal certificate codec.

7.4 Reusable source templates

The template module exposes

```
compile_source_top_tree(triangulation)
verify_source_top_tree(triangulation, template)
specialize_source_top_tree(triangulation, coordinates, geometry, template).
```

The compiled object contains the source fingerprint and an ordered list of vertex, parent, and source-edge triples. Template verification checks parent precedence and exact source edges without invoking the compiler. The specializer emits the same top-trace schema as the queue; the independent top checker therefore accepts it without trusting the template planner.

Templates are an optional API in this release. The default adapter uses the linear queue, which is already inexpensive and chooses rows in the maintained diagnostic's source setting. Choosing a

universal template instead can change the surviving two-handle rows. Its effect on raw completion for genuine geometric inputs with long rows remains to be determined. The uniform validity theorem does not establish downstream optimality. A policy change should consequently be measured with the complete terminal consumer.

7.5 The forest prototype's trust boundary

The abstract planner accepts three-handle labels, cap endpoint occurrence lists, and nonnegative occurrence rows. Missing cap ends mean exterior boundary. The two mappings use the same row labels. Its positive certificate contains a dual forest, an ordered two/three trace, and an ordered one/two unit-column trace. Failure certificates contain a disconnected boundary-free cap component, a forced-row cycle, or a rank-deficiency witness with an optimal residual tree.

The separate verifier recomputes graphic ranks by component traversal and the contraction rank identity. It does not invoke the producer's exchange routine. It checks supplied one/two traces through source occurrence timestamps and supplied two/three traces through endpoint multiplicities. Testing with producer functions disabled verifies this software separation.

These are certificates of the supplied graph cancellation model. Its argument list does not authenticate embedded attaching curves, compatible port collars, or the chain-complex identity. A topological caller must supply those facts. On native inputs the new source top checker provides the cap-side authentication, while the source-parallelity reconstruction provides the attaching cycles. Section 5 explains why this extra structure changes which forest-search claims are meaningful.

7.6 A reviewable integration contract

The package contains a patch against `8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`, an overlay of every changed or added source file, focused tests, raw experiment data, source hashes, and build instructions. Apply the patch or copy the overlay, not both. Incoming packages were inspected but not applied as dependencies. Thus the patch targets the actual maintained baseline rather than an unrecorded mixture of proposals.

A successful integration should preserve the following invariants:

1. Original source proofs authenticate the original geometry before any new terminal transformation.
2. Exact source end occurrences, including double contacts at loops, determine top eligibility.
3. The relative theorem protects the complete native parallelity interface; broader boundary patterns require their own transport contract.
4. An inconclusive raw or restricted-forest result never becomes a negative knot or handlebody verdict.
5. The new helper remains independent of normal multiplicity expansion.

The code is intentionally additive at this boundary. A wider redesign of the hierarchy controller, an integration of the research matroid planner, or a change to the complete recognizer's claimed complexity requires additional work beyond this package.

8 Experiments and what they establish

The experiments serve three different purposes: regression and source-proof checking, controlled local timing, and falsification of overly broad optimization claims. Their scopes should not be mixed. The principal environment was Python 3.12.14 on Linux with Regina 7.4 for optional explicit topological checks. Exact commands, versions, source hashes, raw timing samples, and the saved proof objects are included in the package.

8.1 Whole-cut coverage on the retained corpus

The unchanged corpus is `fast/marked_parallelity_research/corpus.json`, with 106 supplied source/vector cases. Its SHA-256 is recorded in both audit reports. The baseline was copied from the pinned commit before any edit; the modified run used the new top adapter with the same existing terminal policies.

Whole-cut result	Baseline	With top adapter
Complete censuses	85	92
Independently replayed complete censuses	85	92
Inconclusive	21	14
Three-handle exclusion	10	0
Raw cancellation stall	8	10
Raw ball cancellation obstruction	3	4

Table 1: The certified terminal primitive gains seven cases and preserves all 85 previous successes. The redistribution of failure reasons records progress beyond the old three-handle gate. It does not turn remaining refusals into negative topological verdicts.

All ten former three-handle cases receive a complete top reduction. Seven then reach a complete whole-cut census, listed in table 2. The other three remain inconclusive. Fresh Regina metric cuts and handlebody checks were run for all ten and agree with the retained explicit reference data. Each of the three remaining cases contains a nonhandlebody piece according to that explicit check. In the JSON, a genus value of -1 is the oracle's nonhandlebody sentinel, not a negative topological genus.

Newly certified case	Census (genus, multiplicity)
`pilot/interior_1_2/0`	$(1, 2)$
`pilot/interior_2_3/0`	$(1, 2)$
`pilot/interior_3_5/0`	$(1, 1)$
`pilot/finite_trefoil_interior/0`	$(2, 1)$
`interior_1_2/mixed`	$(0, 2)$
`interior_3_5/mixed`	$(0, 2)$
`finite_figureEight_interior/mixed`	$(0, 1), (2, 2)$

Table 2: All seven new version-6 certificates are saved with their exact source/vector inputs and independently replayed. These are censuses of specified cuts, including cuts of nontrivial knot exteriors.

The three remaining former exclusions are `pilot/finite_figureEight_interior/0`, `interior_2_3/mixed`, and `finite_trefoil_interior/mixed`. Among the new total of 92 successes, 85 have

explicit stored-oracle comparisons. The seven very large binary scale cases are checked by independent compressed replay and their retained scale expectations rather than literal expansion. The audit wall times include different checking work and were not collected as a paired speed comparison; they support no overall speedup claim.

8.2 Regression, scale, and adversarial proof checks

Fifty focused integration and regression tests passed: nine new top-handle tests, seven existing whole-cut tests, and 34 existing core, annulus, bundle-kernel, and initial-source-reduction tests. Five further source-tree tests and twelve forest-exchange tests also passed. These are 67 distinct named methods, not a claim to have rerun every historical test in the repository.

The top tests include an actual five-tetrahedron solid torus with an interior loop edge, obtained by a 1–4 and a 2–3 move. Its isomorphism signature is `fLHPccdedehspga`. A forged trace that treats the loop's two contacts as one is rejected. Another source has five interior vertices and a genuinely dependent rooting order; moving a dependent cancellation to the front is rejected. Rank-address changes, source mismatches, repeated IDs, omitted proofs, and schema downgrades are tested. Verifier calls also succeed with the corresponding producer functions disabled.

The reusable template was specialized on all 64 vectors obtained with coefficients $0, 1, 2, 3$ in a compatible sphere, boundary-disk, and Möbius band fixture. All traces independently replayed; two different specialized move choices occur. Separate scale tests at 1, 2^{64}, and 2^{16384} produce top certificates of exactly 193 bytes with the stated JSON serialization, while the largest coordinate needs 16,385 bits. This is evidence for the trace-size statement, not for constant total certificate size.

The abstract forest prototype was compared with exhaustive tree and row checks on 12,350 labelled small graph instances and 1,200 seeded general-row instances. Tests include forced contractions, loops, parallel edges, exact deficiency optimization, malformed rank witnesses, and 16,385-bit labels or counts. These instances test the graph theorem and implementation; most were not produced by three-manifolds.

8.3 A controlled benchmark on actual source triangulations

Starting from the one-tetrahedron layered solid torus, perform q interior 1–4 moves, always choosing the tetrahedron with the fewest boundary corners and then the lowest index. Use the empty normal vector. The resulting actual source has

$$t = 3q + 1, \qquad h_2 = 4q, \qquad h_3 = q.$$

Six sizes, $q = 1, 4, 16, 64, 128, 256$, were measured. Geometry was constructed and independently authenticated outside the scheduler timers. Both the faithful old diagnostic scan and the queue produced top traces that independently replayed at each size.

Each scheduling comparison used nine randomized paired samples, calibrated batches, and an additional nine queue/queue A/A pairs. Five samples measured the common source-incidence reconstruction, and five randomized pairs measured source reconstruction plus scheduling. The scan keeps its original full sorted scan and candidate-list construction. Normalizing its final output is excluded from its timer, while the queue includes survivor output and counters; this choice favours the scan.

q	t	Scan (ms)	Queue (ms)	Local ratio	Combined ratio
1	4	0.00118	0.00434	0.27×	1.01×
4	13	0.00936	0.01670	0.56×	0.94×
16	49	0.16567	0.08763	1.89×	0.98×
64	193	3.04399	0.35683	8.53×	1.20×
128	385	11.20248	0.73123	15.32×	1.46×
256	769	47.09763	1.63552	28.80×	1.94×

Table 3: Ratios are scan median divided by queue median. The combined stage reconstructs source incidence for both methods. It excludes initial parallelity construction, source-proof authentication, and terminal recognition. Values below one are regressions.

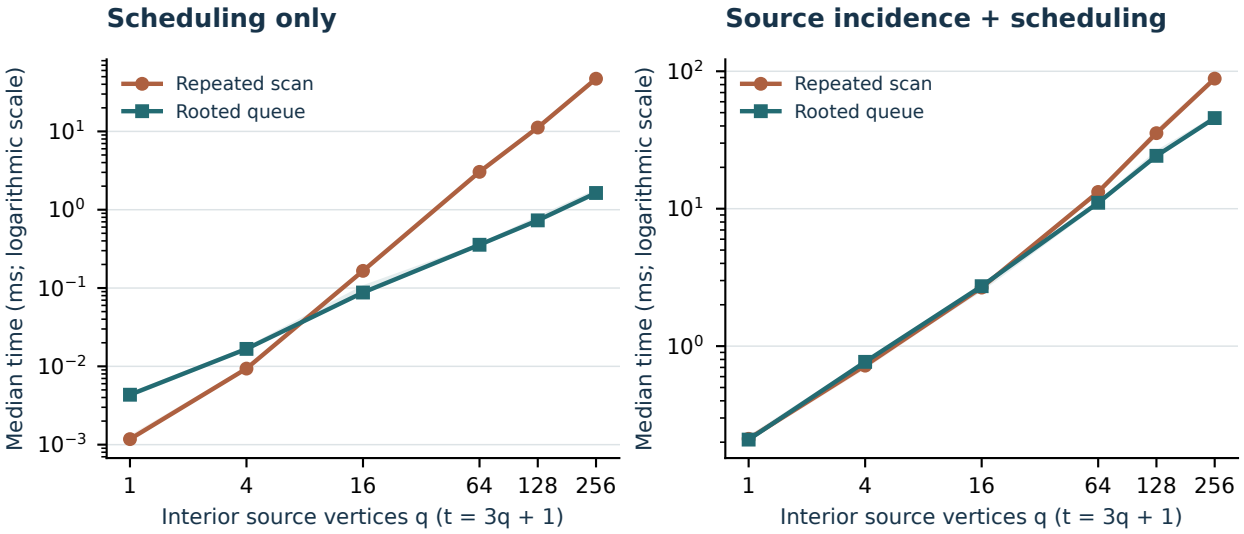


Figure 2: Retained medians with interquartile bands. Common source work substantially dilutes the scheduling gain. Both axes use logarithmic scales; the two panels have different vertical ranges.

At $q = 256$, the isolated medians are 47.098 ms for scanning and 1.636 ms for the queue. Common source-incidence reconstruction alone takes 45.541 ms; the directly measured combined medians are 88.425 ms and 45.655 ms. Thus the observed ratios are 28.80 and 1.94, respectively. The queue has greater constant overhead for the smallest inputs. A/A median paired ratios range from 0.967 to 1.070; the $q = 16$ ratio-of-medians A/A statistic is noisier, 1.189. Small timing differences should not support a strong interpretation. The largest effects are substantially larger than these controls, but still concern a local Python implementation on a shared host.

There is also an exact asymptotic comparison independent of timing. With $m = 4q$, the old loop visits

$$qm - \frac{q(q-1)}{2} = \frac{7q^2 + q}{2}$$

rows. At $q = 256$ this is 229,504 row visits and 457,462 actual interior-cap membership tests. The queue performs 2,038 incidence updates; its explicit upper bound including initialization is 9,426 endpoint inspections. The old scan performs quadratic row-scan work on this genuine source family, while the new queue has the linear structural bound of proposition 3.4.

8.4 The forest-planning experiment gives a negative result

The native applicability audit fixes the maintained primal forest and compares the abstract restricted planner with the ordinary rooted top queue followed by raw core cancellation. On the 106 source cores, the restricted planner completes 40 and refuses 66 because forced long rows already contain a cap cycle. Ordinary core cancellation completes 100. The restriction therefore misses 60 ordinary successes and adds no coverage. On eight subdivided empty solid tori it completes only $q = 1, 2$, while ordinary core cancellation completes all eight. This observed family is not claimed as a theorem about every later subdivision.

These numbers concern the core before all attached bundles are resolved. They are not the 92 whole-cut successes of table 1. In particular, all ten former three-handle cores pass the ordinary core consumer; the three remaining whole-cut refusals arise later.

All 114 source cores in the chain audit satisfy the uncapped $A^{\mathsf{T}}B = 0$ identity; 18 checks involve actual three-handles. The mandatory set $\overline{D}$ is a cap forest in 40 corpus cases and two subdivision cases. Every one has $\beta_2 = 0$, and the contracted rank formula and source trace replay agree. The data do not empirically test nontrivial forest variation: in every eligible case $|\overline{D}| = q$, so the eight proposed edge orders yield the same tree. All full cores in this audit have $\beta_2 = 0$. The invariance and positive-nullity statements are supported by the proof and the explicit geometric family in example 5.4, not by an overstated experimental claim.

8.5 The exceptional-row quotient retains the missing information

The fixed-forest quotient was checked on 21,304 ternary matrices with at most three rows and columns, 4,096 binary 4×3 matrices, 3,375 legal exceptional partitions of binary 3×3 matrices, and 2,000 seeded larger matrices. Tests verify decision equivalence, exact canonical stopping rows, lifted source schedules, ground handling, duplicate types, and large coefficients. Separate replay verifies all 120 retained abstract and native certificates without importing the producer.

For the ordinary fixed dual forest in the native corpus, all 106 short-row graphs are forests. The quotient recovers the 100 complete cores and six canonical stopping cores. In aggregate, 773 source rows reduce to 164 exceptional rows, and 809 occurring source columns reduce to 189 distinct nonzero quotient types. The maximum exceptional-row count is nine, with median one. All eight subdivision cores also complete. These are exact structural compression measurements. Ordinary peeling already decides these explicit matrices, so we claim no new measured runtime or coverage gain from the quotient itself.

9 Consequences for the quasipolynomial programme

9.1 What the local theorem removes from the global problem

The native source top handles need no separate combinatorial search. For every supplied normal surface, a linear trace exists, and a single source tree works uniformly across all its coordinate supports. A hierarchy controller can therefore eliminate this class of three-handles before a terminal core test without branching over possible rooted schedules merely to obtain a valid reduction.

This does not imply that all deletion trees are interchangeable for the later general raw problem. Section 5 proves interchangeability for a specific short-row strategy, while section 6 isolates the additional information carried by long rows. The next useful search target is consequently the inter-

action of exceptional attaching curves with forest components, rather than unrestricted enumeration of short-row forests that homology already controls.

Proposition 9.1 (Conditional preservation of a global bound)**.** *Suppose a correct complete hierarchy algorithm on an n-crossing input visits at most $N(n)$ source states, each with encoded size at most $L(n)$. Suppose the required source verification, top reduction, and selected terminal operations at one state cost at most $L(n)^c$, for a fixed c. Then these operations contribute at most $O(N(n)L(n)^c)$ bit work. If both N and L are quasipolynomial in n, their contribution remains quasipolynomial. The source top reducer adds no coordinate-support search factor to this estimate.*

Proof. Sum the stated per-state cost over the visited states. If $N(n) \le \exp(O((\log(n+2))^a))$ and $L(n) \le \exp(O((\log(n+2))^b))$, their product is $\exp(O((\log(n+2))^{\max(a,b)}))$. The uniform source-tree theorem supplies one valid top schedule without an enumeration of supports or trees. □

This is accounting, not a proof of the missing hypotheses. In particular, polynomial verification of a supplied state does not bound the number of candidate states, and a short hierarchy does not bound the time needed to select its next admissible decomposing surface. The present work does not establish either N or L for a complete recognizer.

9.2 Immediate integration and performance questions

Research question 9.2 (Reuse authenticated source work)**.** Can one immutable validated-source context serve the parallelity, boundary, top-incidence, and terminal checks without repeating the same triangulation and coordinate reconstruction?

The largest local benchmark spends more time on common incidence reconstruction than on the new queue. This is a concrete next performance target. A suitable design would expose source invariants through an internally constructed object, retain exact input identity, and make untrusted serialized objects pass the full constructor. The producer and verifier should still have separate geometric reconstructions where independence is part of the certificate contract. A successful change should report whole-query timings, retained memory, and invalid-input rejection, not only cache-hit speed.

Research question 9.3 (Choose a stable top policy without losing terminal cases)**.** When does the universal source-tree policy give the same terminal success set as the queue, and when does a different valid top tree help with exceptional rows?

The template proves validity and makes caching possible. It does not prove optimality after the rows change. A useful experiment should compare policies on actual source cuts, retain the full attaching records, and separate short-row cases governed by homology from long-row cases where a choice could matter. An abstract graph example violating the chain identity is not suitable evidence. The minimal-subtree stopping lift provides a specific geometric region to inspect when a policy differs.

Research question 9.4 (Extend the kernel to the complete finite terminal matrix)**.** Can the exceptional-row quotient be compiled directly on the combined core and saturated-bundle boundary kernel, with the original geometric projection and private-cocore accounting preserved?

The current quotient experiment covers cores. The seven new whole-cut successes show that the top adapter already composes with several existing bundle terminal paths, but the remaining obstructions occur in the full assembly. The next adapter should retain row ownership, complete interface circles, and the proof that omitted private columns cannot become valid pivots. It should then lift a quotient stopping set to the corresponding source annuli and core attaching curves. This would produce a better description of each unresolved whole cut, without claiming that the description itself resolves it.

9.3 Geometric questions that could change the asymptotics

Research question 9.5 (Transport a full inherited boundary pattern)**.** Can the two/three-handle collar homeomorphisms be encoded and composed in polynomial bit size while transporting arbitrary admissible pattern curves and trivalent graphs?

For the current parallelity ports, support avoidance makes the relative map simple. A general pattern may cross the support. One possible approach is to retain local PL boundary maps as a composition grammar, with source flags and affine rank intervals on each patch. The hard part is then an exact intersection and admissibility checker that operates on the grammar without expanding its maps. The needed theorem must bound both the grammar and the complexity of the queries used by the next hierarchy step. A short list of moves alone does not bound the expanded boundary pattern.

Research question 9.6 (Exploit few exceptional rows)**.** For a geometrically authenticated state with k exceptional rows, can a complete useful family of forest changes or local geometric repairs be searched in $(t + b_x)^{O(k+1)}$ time, or with another bound that becomes quasipolynomial when k is logarithmic?

This is a proposed target, not a conjecture supported by a proof. The fixed-forest quotient gives an exact state description and at most $3^k - 1$ nonzero capped column types. It does not show that the set of all such descriptions reachable by forest changes is small, nor that local changes preserve completeness of a search. A serious attempt should first seek a counterexample to any claimed exchange axiom or locality bound. Proposition 4.7 already rules out the simplest unrestricted matroid formulation.

Research question 9.7 (Change the primal forest and cocore system)**.** Can a primal zero/one-handle forest be chosen so that the number or total interaction complexity of the exceptional rows is provably small?

The homological rigidity theorem fixes that forest. Changing it changes the surviving columns and can change which attaching curves are short. This is a more plausible source of useful freedom than short-row dual-tree exchange. The exact parameter to minimize should incorporate geometry: the number of long rows alone ignores the number and placement of their pin columns in the quotient. A proposed optimization should retain a certificate for the cocore correspondence and should not substitute free group exponent reduction for geometric intersection reduction.

Research question 9.8 (Destroy a lifted stopping core by controlled geometric moves)**.** Which handle slides, disk exchanges, or boundary bigon removals can reduce the unsigned intersection data near a lifted stopping set while preserving the marked manifold and a useful global complexity measure?

The exact lift identifies the exceptional rows and the short-tree paths that keep them blocked. It therefore localizes a candidate repair region. It does not prove that a necessary repair is confined there. The desired statement would give a complete family of repairs, a compact proof for each move, and a potential that decreases by a controlled amount. A counterexample where all useful repairs leave this region would be equally valuable, because it would rule out an attractive but incomplete search.

Research question 9.9 (Batch compressed primitive reductions)**.** Can large families of equivalent curve slides be accelerated in the compressed source coordinates, with an exact geometric certificate for their combined effect?

Compressed free-group primitivity is a relevant comparison, but it is not a replacement for embedded curve data. Kapovich's 2026 work distinguishes rank-two polynomial algorithms from more general compressed settings and gives the family ab^{2^n} as a warning about exponentially many elementary Whitehead descent steps [5]. A geometrically useful batching theorem must specify how an accelerated algebraic step acts on the actual boundary curves, their disjointness, and the protected pattern. Large binary coordinates make batched operations attractive precisely because expanding the implied elementary moves defeats the objective.

9.4 Global control and verification questions

Research question 9.10 (Find a stable hierarchy potential)**.** Is there a potential on authenticated marked source states whose decrease controls both useful-surface selection and the cost of rebuilding the compressed representation?

Removing top handles changes two handle counts equally and preserves Euler characteristic. It is not by itself a strict decrease of a global topological complexity. The potential must account for source size, coordinate bit length, pattern complexity, and any reset that invalidates cached geometry. A bound on the number of moves inside a phase is insufficient if the next phase rebuilds a much larger state. A candidate proof should state a recurrence for these quantities together, rather than prove separate bounds that depend circularly on one another.

Research question 9.11 (Control useful surface discovery)**.** Can the existing separator, first-return, and surface-batch proposals be connected to an authenticated complete choice of the next decomposing surface, with explicit bit bounds?

The incoming reports provide several useful compressed graph and normal coordinate operators. Their output graphs, surfaces, and retained marks must still be connected to the admissibility and essentiality conditions of a complete hierarchy. The source top theorem can be used after such a choice; it does not make the choice complete. A deliverable at this stage should include both a source exporter and a negative certificate describing what exhaustive failure means in the chosen restricted problem.

Research question 9.12 (Make the source interface reentrant)**.** Can later hierarchy states with several boundary components be accepted by a generalized source constructor without losing the topology and pattern checks now guaranteed at the one-torus entry point?

The relative top theorem already permits several nonempty boundary components. The native validator does not. Generalizing it is more than removing the torus test: all consumers of boundary

component counts, compressing disks, source provenance, and essential-pattern criteria must be given explicit contracts. This is a natural point to connect the initial knot-diagram provenance certificate to the sequence of derived states.

Research question 9.13 (Mechanize the small certificate kernel)**.** Which of the cap inventory, rooted trace, graphic peeling, and quotient stopping-lift lemmas can be formalized first, while keeping the required PL geometric theorem explicit?

The finite combinatorial checker is small enough to make formalization realistic. A useful first milestone is a theorem that every accepted trace has the exact graph and row consequences stated here. The source geometry then remains a clearly named theorem connecting that finite kernel to embedded handles. Python tests and independent implementations provide strong regression evidence but are not machine-checked proofs of the PL statement.

The highest-value next steps are the source-context optimization, the full terminal stopping lift, and a marked geometric repair operation with a proved completeness or progress property. They address observed costs or specific unresolved geometry. Another abstract forest optimizer without the chain and embedding constraints would not address the main barrier identified by this work.

A Reproduction and artifact map

A.1 Pin the input before applying the change

The integration patch is relative to the ProveIt repository root. Its target commit is

`8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`.

Its manifest records the SHA-256 of every supplied file and the original hash of each modified file. The package does not include or apply any of the ten incoming ZIP patches.

For a clean comparison, retain one checkout at that commit and apply `integration/proveit-relative-top-handles.patch` to another. The command `git apply -check` should succeed before applying it. The parallel `integration/overlay/` tree is a convenient full-file alternative. It is not an additional patch to apply afterwards.

A.2 Run the focused release checks

Set `PACK` to the absolute unpacked package path and `FAST` to the target repository's `Topology/UnknotRecognition/fast` path. Then run:

```
python "$PACK/tools/run_targeted_tests.py" \
  --fast-dir "$FAST" --output-dir /tmp/top-release-check

cd "$FAST"
python core_top_research/audit.py \
  --fast-dir "$FAST" \
  --reference ../synthesis/data/boundary-kernel-audit.json \
  --oracle-top --output /tmp/extended-audit.json \
  --save-certificates /tmp/new-top-certificates
```

The test runner selects the eight named focused modules and writes a full log plus machine-readable counts. All 67 methods pass in the separate clean patch application. The runner also

requires the unchanged source fixtures `synthesis/data/core-three-handle-probe.json`, `synthesis/data/handle-reducible-trefoil.json`, and `reports/57/results/discovery_corpus.json`, relative to `Topology/UnknotRecognition/`; it checks these before starting. The corpus audit requires the retained reference JSON and compares all 106 cases; `-oracle-top` uses Regina for fresh explicit cuts on the ten former exclusions. Omit that flag when only the native source replay is required. These fresh explicit checks can be more expensive than the compressed native operation.

The current audit script can also test a baseline checkout by pointing `-fast-dir` at its unmodified fast directory and omitting `-oracle-top`. The script itself must be invoked from the new package or overlay, since it is not part of the pinned baseline.

A.3 Reproduce local and structural experiments

From the patched fast directory:

```
python core_top_research/benchmark.py \
  --output /tmp/top-scheduler-benchmark.json
python core_top_research/template_audit.py \
  --output /tmp/template-audit.json
python core_top_research/forest_audit.py \
  --output /tmp/forest-exchange-audit.json
python core_top_research/chain_audit.py \
  --subdivision-source /tmp/forest-exchange-audit.json \
  --output /tmp/chain-compatibility-audit.json
cp /tmp/forest-exchange-audit.json \
  core_top_research/forest_exchange_audit.json
python core_top_research/residual_kernel_audit.py \
  --output /tmp/residual-kernel-audit.json
python core_top_research/residual_kernel_audit.py \
  --replay-only "$PACK/evidence/residual_kernel_audit.json"
```

The default benchmark uses the reported six sizes, nine scheduler/A–A pairs and five combined-stage pairs. Hardware and load changes can alter timings; the exact work counts, source hashes, and proof consequences are the reproducible mathematical comparisons. The residual-kernel audit recreates its finite matrix tests and the native corpus. Its subdivision fixtures are read from the required neighbouring file `core_top_research/forest_exchange_audit.json`; the copy command above supplies that file. Alternatively, copy the archive's retained `evidence/forest_exchange_audit.json` to that location before the kernel audit. The replay-only command reads its supplied JSON directly.

A.4 Build the article and inspect the evidence

All TeX sections and the vector timing figure are included. From `article/`, build with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  Relative_Top_Handles.tex
```

The article uses standard TeX Live packages. The optional `tools/render_evidence.py` regenerates the timing table and figure from the retained JSON with Matplotlib. It does not run a new measurement.

The key evidence files are:

File under `evidence/`	Contents
`baseline_audit.json`, `extended_audit.json`	Per-case results, source corpus hash, proof sizes, and oracle comparisons
`new_certificates/`	Exact source/vector inputs and all seven new whole-cut certificates
`top_scheduler_benchmark.json`	Six real source triangulations, both traces, operation counts, raw paired samples, A/A controls, and stage timings
`template_audit.json`	Fixed source tree, 64 compatible vectors, and binary-scale proofs
`forest_exchange_evidence.json`	Abstract planner examples, independent replays, and exhaustive-test summary
`forest_exchange_audit.json`	Native restricted-strategy results and eight subdivision fixtures
`chain_compatibility_audit.json`	Uncapped source matrices, chain checks, ranks, mandatory cycles, and traces
`residual_kernel_audit.json`	Quotient matrices, source lifts, exact stopping rows, and retained replay cases

A.5 Claim boundaries for later integration

The article gives mathematical proofs, and the package gives independent programmatic replay and regression evidence. It is not a formal proof assistant development. The abstract failure witnesses establish a failure of a named matrix strategy. The native whole-cut successes establish unmarked handlebody censuses for specified cuts. Neither is automatically a final essential-boundary-pattern certificate or an unknot verdict.

The archive's claim ledger and integration notes repeat these contracts so that a future importer need not infer them from benchmark filenames. All third-party sources are referenced by public links; their full papers or lecture slides are not redistributed in this package.

B Repository and incoming-work audit

The review used the maintained source at the pinned commit and the ten ZIP files then present under `docs/incoming`. The incoming audit covered READMEs, abstracts, relevant theorem statements and proofs, implementation boundaries, and research agendas. It is a targeted overlap audit, not a claim that every proof in every imported manuscript was independently verified. File hashes and the more detailed audit notes are included with the deliverables.

Incoming topic	Relevant existing result and boundary of overlap
Cographic Persistence	Surface-batch independence, cographic optimization, and rebuild accounting. Its matroid concerns retaining decomposing surfaces, not selecting source two/three-handle deletion trees.
Disconnected Tile Kernels	Compressed graph operators for disconnected repeated tiles, with contextual replacement. It does not authenticate source top-handle geometry.

Incoming topic	Relevant existing result and boundary of overlap
Genus Budgets and Native Interfaces	Initial source handle compilation and reduction, genus budgets, and native interface data. The initial two/three queue is credited; the later normal-cut adapter with preserved ports is the extension here.
Modular Morse Certificates	Verification of supplied interval matchings and relative product lifts. No general matching discovery or native exceptional-three-handle adapter is supplied.
Moment-Coded First Returns	Weighted orbit summaries, transport, and first-return compression. These are source data operators, not the relative top-handle move proved here.
Saturated Bundle Cancellation	Geometric residual relations for saturated bundles, private-cocore projection, and 85/106 native censuses. The unsupported three-handles are an explicit remaining problem.
Scalar Separators	Exact restricted normal-coordinate message rules and a separator obstruction. This concerns surface discovery, not top-handle elimination.
Unknot Hierarchy Kernels	Relative bands for nonsaturated bundles, bounded core kernels, and repeated-component graph reductions. Saturated relations and three-handles remain distinct obligations.
Sparse Cut Grammars	Exact compressed graph cut responses and contextual interfaces. A faithful geometric source exporter and embedded noose transport remain necessary.
Stopping Cores	Complete unit-column characterization of positive raw cancellation, canonical stopping sets, passenger transport, and native saturated tests. Its completeness theorem is reused; the cap-forest chain constraint and exceptional short-tree quotient are new relative to this inspected work.

The maintained `synthesis/data/core-three-handle-probe.py` already computes candidate cap endpoints, removes rooted pairs, and predicts seven successful downstream cases. Its own stated limitation is missing independent source replay. We therefore do not present the diagnostic's initial idea or its seven predictions as newly discovered. The advance is the relative proof, orientation-safe reconstruction, uniform source template, strict compositional certificate, and demonstrated integration.

The source core bounds and rank charts come from the maintained `normal_parallelity_geometry.py` and its independent `normal_parallelity_reference.py`. The geometric occurrence matrix is read from complete attaching-cycle flags. The prior initial source queue in `source_handle_reduction.py` confirms that boundary-rooted cancellation is already part of the project's vocabulary; it did not by itself justify the new relative normal-cut use.

Novelty is consequently claimed at the level supported by this audit: specific source-relative theorems and adapters, the chain-aware analysis of the proposed forest optimization, and the exact exceptional-row stopping quotient. Classical handle cancellation, graphic matroids, matroid partition, and stopping-set theory are acknowledged as underlying tools. No assertion is made that the full literature contains no equivalent formulation of any elementary graph lemma.

References

[1] Marc Lackenby, *Incompressible surfaces, hierarchies and unknot recognition*, arXiv:2607.23350v1, 25 July 2026. In particular Sections 9–10 and Theorem 14.1. https://arxiv.org/abs/2607.23350v1.

[2] Marc Lackenby, *Unknot recognition in quasi-polynomial time*, February 2021 lecture slides, 109 pages; announcement on PDF pp. 13–14. https://people.maths.ox.ac.uk/lackenby/quasipolynomial-talk.pdf. The later Oxford slide version is distinct; the cited bound here is from this February version. Slides are referenced, not redistributed.

[3] Adele Jackson, *Minimal triangulation size of Seifert fibered spaces with boundary*, arXiv:2301.02085v1, 5 January 2023. Lemma 3.2 and the following parallelity discussion, pp. 6–7. https://arxiv.org/abs/2301.02085v1.

[4] Jack Edmonds and D. R. Fulkerson, *Transversals and matroid partition*, Journal of Research of the National Bureau of Standards, Section B, 69B(3) (1965), 147–153. Theorem 1c, constructive exchange proof, and contraction formula. https://doi.org/10.6028/jres.069B.016. Original publication: https://nvlpubs.nist.gov/nistpubs/jres/69B/jresv69Bn3p147_A1b.pdf.

[5] Ilya Kapovich, *Compressed primitivity problem in free groups*, arXiv:2607.21499v1, July 2026. Theorems 1.1–1.2 and Example 1.3. https://arxiv.org/abs/2607.21499v1.

[6] Vladimir Reshetnikov and ProveIt contributors, *ProveIt: Topology/UnknotRecognition*, repository snapshot `8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4`, inspected 12 October 2026. The maintained source, synthesis reports, retained corpus, and diagnostic script are identified in the text and package manifest. https://github.com/VladimirReshetnikov/ProveIt/tree/8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4/Topology/UnknotRecognition.

[7] ProveIt incoming research package, *Stopping Cores*, 12 October 2026. `docs/incoming/ProveIt_Unknot_Stopping_Cores_2026-10-12.zip` at the snapshot in [6]. Used for stopping-set persistence, completeness of positive raw cancellation, and geometric projection scope.

[8] ProveIt incoming research package, *Saturated Bundle Cancellation*, 12 October 2026. `docs/incoming/ProveIt_Saturated_Bundle_Cancellation_2026-10-12.zip` at the snapshot in [6]. Used for source annulus charts, residual bundle relations, private-cocore projection, and overlap analysis.

[9] ProveIt incoming research package, *Genus Budgets and Native Interfaces*, 12 October 2026. `docs/incoming/ProveIt_Genus_Budgets_Native_Interfaces_2026-10-12.zip` at the snapshot in [6]. Used for initial source reduction and interface compilation context.

[10] ProveIt incoming research collection, 12 October 2026. The ten package titles and exact overlap boundaries are listed in appendix B; their hashes are retained in the package audit. https://github.com/VladimirReshetnikov/ProveIt/tree/8cb7d9d25ac8cb64a9d147304e3441e98e94c7a4/docs/incoming.